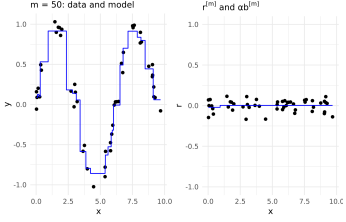
# Introduction to Machine Learning

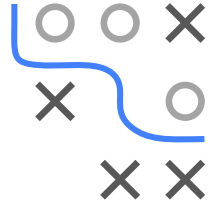
# Boosting
# Gradient Boosting with Trees 1




**Learning goals**

- Examples for GB with trees
- Understand relationship between model structure and interaction depth

# GRADIENT BOOSTING WITH TREES



Trees are most popular BLs in GB.

**Reminder: advantages of trees**

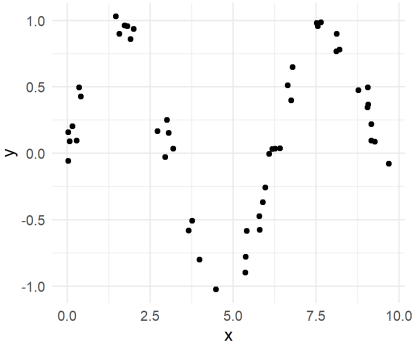
- No problems with categorical features.
- No problems with outliers in feature values.
- No problems with missing values.
- No problems with monotone transformations of features.
- Trees (and stumps!) can be fitted quickly, even for large $n$.
- Trees have a simple, built-in type of variable selection.

GB with trees inherits these, and strongly improves predictive power.

# EXAMPLE 1

**Simulation setting:**

- Given: one feature $x$ and one numeric target variable $y$ of 50 observations.
- $x$ is uniformly distributed between 0 and 10.
- $y$ depends on $x$ as follows: $y^{(i)} = \sin\left(x^{(i)}\right) + \epsilon^{(i)}$ with $\epsilon^{(i)} \sim \mathcal{N}(0, 0.01)$, $\forall i \in \{1, \ldots, 50\}$.



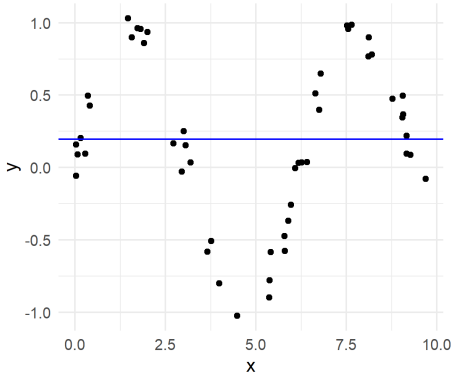

**Aim:** we want to fit a gradient boosting model to the data by using stumps as base learners.

Since we are facing a regression problem, we use $L2$ loss.

# EXAMPLE 1 / 2

**Iteration 0:** initialization by optimal constant (mean) prediction $\hat{f}^{[0](i)}(x) = \bar{y} \approx 0.2$.

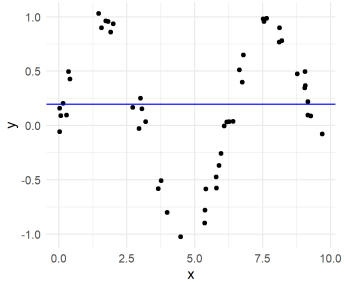
| $i$ | $x^{(i)}$ | $y^{(i)}$ | $\hat{f}^{[0]}$ |
|---|---|---|---|
| 1 | 0.03 | 0.16 | 0.20 |
| 2 | 0.03 | -0.06 | 0.20 |
| 3 | 0.07 | 0.09 | 0.20 |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |
| 50 | 9.69 | -0.08 | 0.20 |



# EXAMPLE 1 / 3

**Iteration 1:** (1) Calculate pseudo-residuals $\tilde{r}^{[m](i)}$ and (2) fit a regression stump $b^{[m]}$.

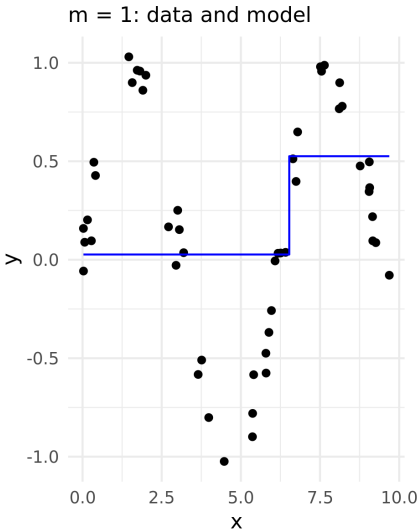
| $i$ | $x^{(i)}$ | $y^{(i)}$ | $\hat{f}^{[0]}$ | $\tilde{r}^{[1](i)}$ | $\hat{b}^{[1](i)}$ |
|---|---|---|---|---|---|
| 1 | 0.03 | 0.16 | 0.20 | -0.04 | -0.17 |
| 2 | 0.03 | -0.06 | 0.20 | -0.25 | -0.17 |
| 3 | 0.07 | 0.09 | 0.20 | -0.11 | -0.17 |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |
| 50 | 9.69 | -0.08 | 0.20 | -0.27 | 0.33 |

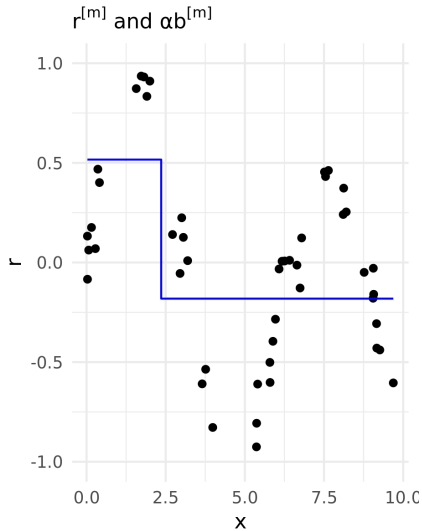



(3) Update model by $\hat{f}^{[1]}(x) = \hat{f}^{[0]}(x) + \hat{b}^{[1]}$.

# EXAMPLE 1

Repeat step (1) to (3):



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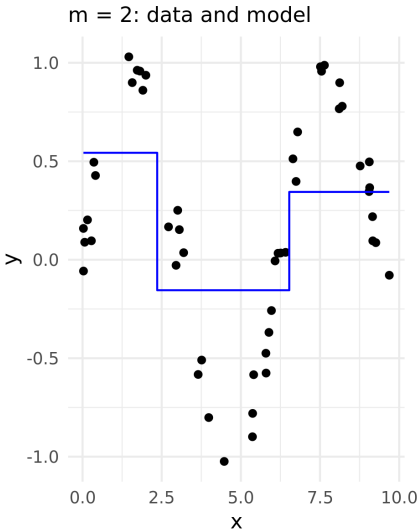


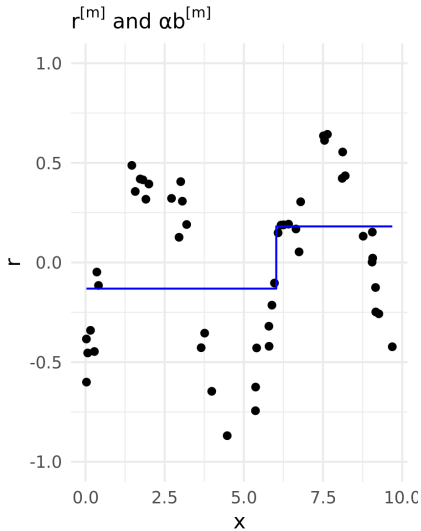

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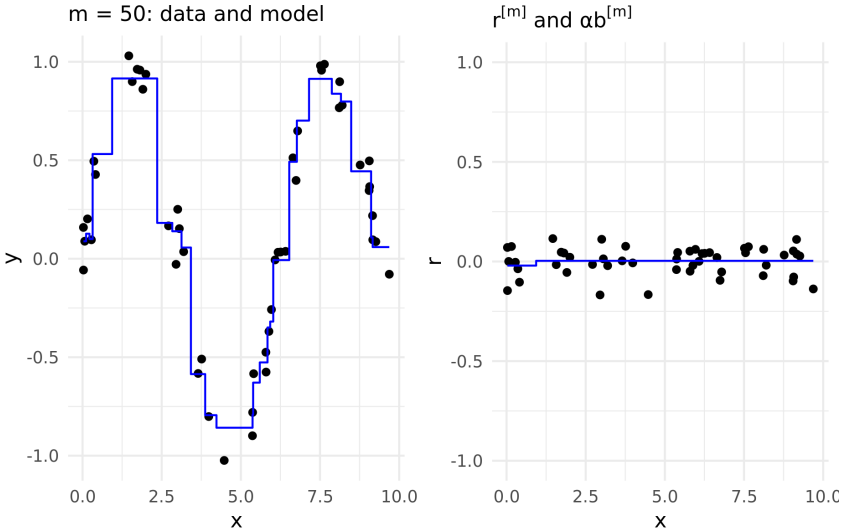
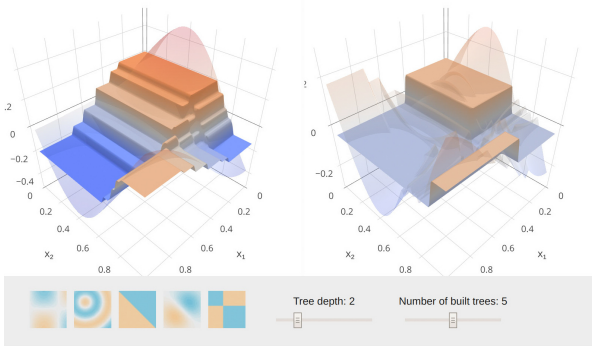
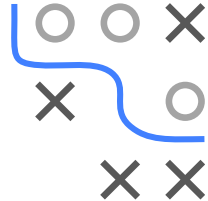


# EXAMPLE 2

This website shows on various 3D examples how tree depth and number of iterations influence the model fit of a GBM with trees.



# MODEL STRUCTURE AND INTERACTION DEPTH



Model structure directly influenced by depth of $b^{[m]}(\mathbf{x})$.

$$f(\mathbf{x}) = \sum_{m=1}^{M} \alpha^{[m]} b^{[m]}(\mathbf{x})$$

Remember how we can write trees as additive model over paths to leafs.

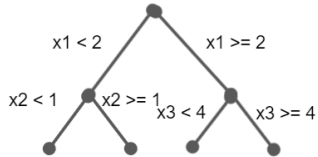
With stumps (depth = 1), $f(\mathbf{x})$ is additive model (GAM) without interactions:

$$f(\mathbf{x}) = f_0 + \sum_{j=1}^{p} f_j(x_j)$$

With trees of depth 2, we have two-way interactions:

$$f(\mathbf{x}) = f_0 + \sum_{j=1}^{p} f_j(x_j) + \sum_{j \neq k} f_{j,k}(x_j, x_k)$$

with $f_0$ being a constant intercept.

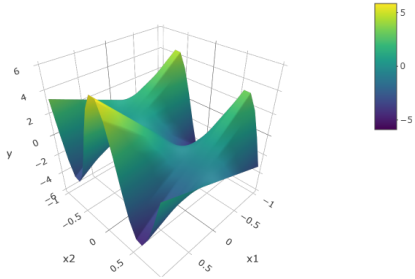


# MODEL STRUCTURE AND INTERACTION DEPTH

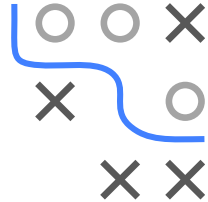
/ 2

**Simulation setting:**

- Features $x_1$ and $x_2$ and numeric $y$; with $n = 500$
- $x_1$ and $x_2$ are uniformly distributed between -1 and 1
- $y^{(i)} = x_1^{(i)} - x_2^{(i)} + 5\cos(5x_2^{(i)}) \cdot x_1^{(i)} + \epsilon^{(i)}$ with $\epsilon^{(i)} \sim \mathcal{N}(0, 1)$
- We fit 2 GB models, with tree depth 1 and 2, respectively.

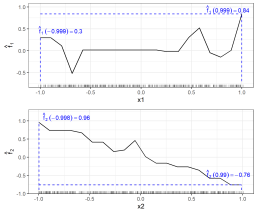


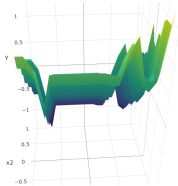
# MODEL STRUCTURE AND INTERACTION DEPTH / 3

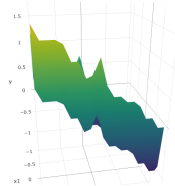


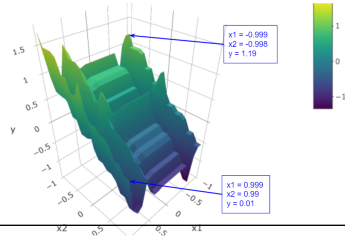
**GBM with interaction depth of 1 (GAM)**

No interactions are modelled: Marginal effects of $x_1$ and $x_2$ add up to joint effect (plus the constant intercept $\hat{f}_0 = -0.07$).








$$\hat{f}(-0.999, -0.998)$$
$$= \hat{f}_0 + \hat{f}_1(-0.999) + \hat{f}_2(-0.998)$$
$$= -0.07 + 0.3 + 0.96 = 1.19$$

# MODEL STRUCTURE AND INTERACTION DEPTH

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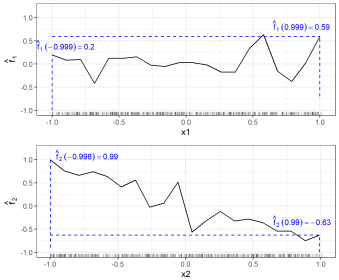
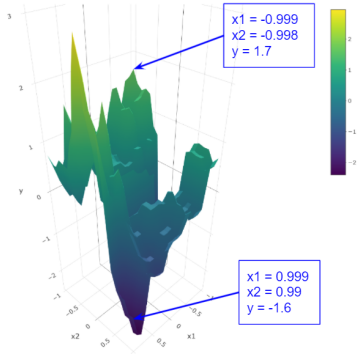
**GBM with interaction depth of 2**

Interactions between $x_1$ and $x_2$ are modelled: Marginal effects of $x_1$ and $x_2$ do NOT add up to joint effect due to interaction effects.